MINI PROJECT: SALES DATA CLEANING

Input file: Messy_data.csv

Output file: cleaned_sales_dataset.csv

Dataset Overview

- Original dataset: 185 rows and 9 columns.
- Exact duplicate rows removed: 5.
- Cleaned dataset: 180 rows and 10 columns (includes TotalAmount).

Cleaning Steps

1. Inspected the dataset using head(), info(), shape and isnull().sum().
2. Removed exact duplicate records.
3. Removed leading/trailing spaces and standardized customer names, cities, products and order statuses.
4. Converted Quantity, UnitPrice and CustomerAge to numeric values. Interpreted 'three' as 3, '2 pcs' as 2, 'forty' as 40 and removed common formatting issues from prices.
5. Treated clearly invalid/extreme values as missing: negative or unusually large quantities, ages outside 18–100, and the extreme UnitPrice value 999999.
6. Parsed mixed date formats. Invalid dates were treated as missing.
7. Filled missing customer names with 'Unknown Customer'; filled missing categorical values with the mode; filled missing numeric values with the median; filled invalid/missing dates with the mode date.
8. Converted columns to suitable data types.
9. Created TotalAmount = Quantity × UnitPrice.
10. Exported the result as cleaned_sales_dataset.csv.

Missing Values and Outliers

- Median was used for numeric columns because it is less affected by extreme values than the mean.
- Mode was used for categorical fields and dates.
- Missing customer names were labeled 'Unknown Customer' rather than assigning an invented name.
- Obvious invalid/extreme entries were treated as missing before imputation. The thresholds are based on the patterns and examples in this teaching dataset and should be confirmed with business rules in a real sales system.

Final Validation

- Remaining missing values: 0
- Remaining exact duplicate rows: 0
- TotalAmount column created successfully.